

From Boxes to Blueprints: An Investigation of User Expectations and Design Strategies for Spatial Augmented Reality UX/UI

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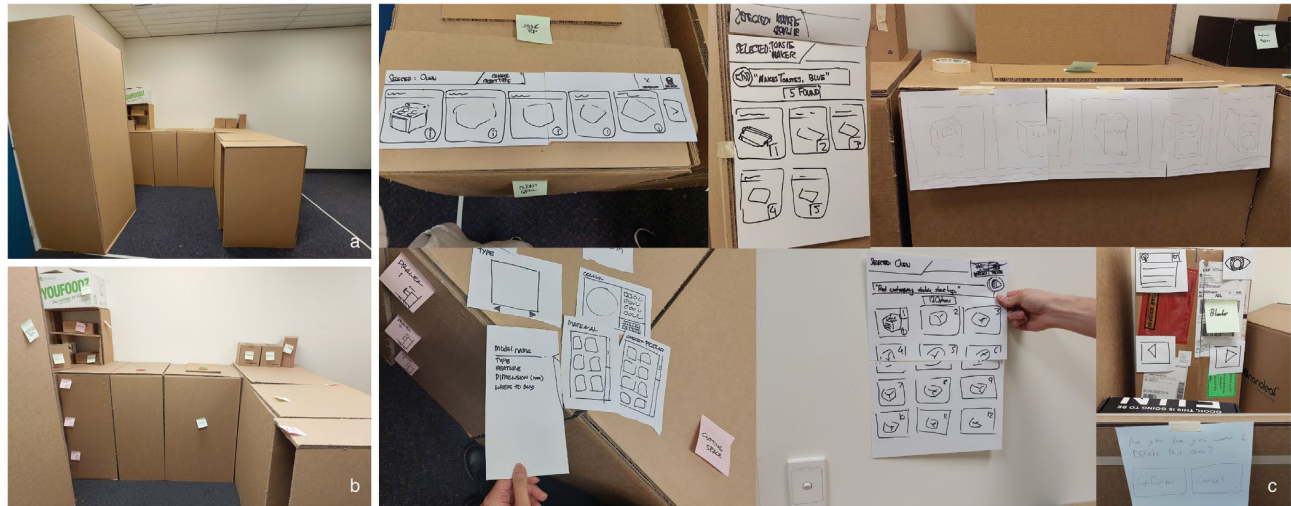


Figure 1: Constructing a small kitchen prototype (a), labelling the boxes (b) and designing SAR UI using cardboard boxes, paper, and pen (c).

ABSTRACT

Designing complex buildings requires certain precisions to assess real-world feasibility and align stakeholders' expectations. Existing design tools often fail to convey scale and physical presence and are limited by perceived inaccuracies, low visual fidelity, and high costs. This research combines cardboard boxes with current Spatial Augmented Reality (SAR) setups for rapid prototyping, where boxes serve as life-size, physical representations that can be overlaid with digital information to enable real-time design modifications. A thematic study was conducted with fifteen participants involved tasks such as arranging physical boxes to prototype a kitchen layout, labelling the boxes, and conceptualizing SAR interfaces and experiences without considering SAR's technical limitations. Key findings highlight touch-based interfaces and gesture controls, especially with visual cues, as the preferred input methods, optimal projection locations on the front and top of boxes and walls, and carousel-style user interfaces as the most

favored design. Further insights contribute toward the development of a comprehensive framework for ubiquitous spatial computing.

Index terms: Human Computer Interaction; User interfaces; Spatial Augmented Reality; World building; Rapid prototype.

1 INTRODUCTION

The concept of “world-building” is used to describe the crafting of detailed settings or spaces by arranging physical elements to shape and define a specific environment. Projection-based Spatial Augmented Reality (SAR) supports large-space design without requiring wearable devices, enhancing user comfort and enabling wider fields of view for collaborative design [1] [2]. The term “world” signifies the versatility of SAR, which can prototype any space. In this presented work, the kitchen context was selected out of the many worlds due to its familiarity and relevance to all participants. SAR was explored as a world-building system, designed to construct spaces and enhance collaboration and communication. It enables users to visualize their ideas, share them with stakeholders, and implement real-time modifications until consensus is achieved.

Complex environments such as hospitals, submarines, and residential spaces demand substantial investments, with construction and correction costs frequently reaching billions of dollars. Design flaws incur significant financial losses and lead to inefficiencies and delays. This highlights a critical opportunity for applications that facilitate creating and visualizing of life-size physical environments augmented with digital content. By incorporating SAR into the early stages of design, stakeholders can align their understanding, rapidly prototype designs for evaluation,

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and easily adjust. This approach minimizes errors, reduces sunk costs, and streamlines the development of complex spaces.

This research speculates how SAR could be utilized as part of designing dynamic environments, leveraging physical cardboard boxes as building blocks for rapid prototyping. These low-cost materials approximate the dimensions and spatial volume of real objects, while SAR overlays virtual appearances onto them, blending physical and digital elements to rapidly create collaborative immersive spaces. This approach fosters intuitive, user-centered interactions and eliminates the constraints of wearables. A user study was conducted to explore how users construct models with cardboard boxes and expect to interact with SAR. The study's goal is to derive design principles emphasizing flexibility, spatial coherence, and task efficiency to inform SAR UX/UI design. It concentrated on life-sized world-building systems, examining UI size, projection surfaces, and mechanisms for selecting, arranging, and configuring digital content within physical boundaries. Through participant surveys, the research identifies trends, behaviors, and deviations to accommodate diverse user preferences. The primary objective was on identifying usage challenges and contextual problems, rather than assessing technological development.

2 RELATED WORK

The affordances of SAR render it particularly well-suited for applications in industrial design tasks [1]. SAR offers an abstract level allowing meaningful alterations in world building. For example, SAR can transform a box's material by projecting textures onto its surfaces. This research aims to explore SAR's dynamics in world-building, focusing on understanding how the user may expect to leverage SAR during the design ideation stage [2].

There have been multiple improvements in Spatial Augmented Reality (SAR) technology, aiming to achieve the concept of spatial prototyping alongside evolving methods for user-system interaction. It began with John Underkoffler's 1997 introduction of Anti-sedentary Beigeless Computing [3], which rejected traditional human-computer interaction and led to the development of the Luminous Room. This project integrated digital interaction within architectural spaces, forming a foundation for future innovations [4] [5]. Building on this concept, the Office of the Future [6] introduced immersive augmented reality experiences, while Life-Sized Projector-Based Dioramas [7] employed projection systems to simulate realistic environments, focusing on surrounding area customization rather than internal elements. Subsequent advancements included RoomAlive [8], a projection mapping prototype enabling interactive digital content in physical spaces, and Dyadic projection [9], which explored collaborative SAR and multi-perspective interactivity. Both systems faced challenges with projection distortion from different viewpoints; however, these issues can be mitigated through proper calibration approaches [10] [11] [12] [13].

Commercial systems such as CADwalk¹ and BigPlans², project floor plans provide accessibility features, enabling users, including those with physical disabilities, to navigate life-sized CAD models [14]. These systems support rapid spatial prototyping across industries such as healthcare and logistics. Other notable works include Illuminating Clay (real-time landscape modeling) [15], Peter Schumacher's life-sized hospital prototypes [16], and Huawei³'s 3D smart home mapping installation [17].

3 USER STUDY

This study investigated SAR's potential to enhance spatial design by integrating projection technology with physical props for rapid evaluation of building designs. Fifteen participants, recruited from the general public and university, mostly without design or construction experience, engaged in a kitchen design task, leveraging SAR to design physical spaces. Despite SAR's limitations [18], the research emphasized the multimodal integration of projection technology with other systems to support intuitive, immersive design. The study received ethics approval, and participants were compensated \$20 AUD for their time.

The cardboard boxes in this study are categorized into two groups: large boxes with standardized sizing and miscellaneous-sized boxes. IKEA⁴, a globally recognized company known for its consistent furniture measurements and publicly available product specifications, serves as a useful reference [19]. The dimensions of IKEA's furniture will be used as the standard for the boxes in this study. Categories include X-Small (250x630x930 mm), Small (400x630x930 mm), Medium (600x630x930 mm), Large (900x630x930 mm), and Fridge (595x630x1874 mm). The study's kitchen area was 3.1x1.2m.

1.1 Methodology

This study employed an observational and thematic approach to explore participants' behaviors and preferences when engaging with a hypothetical SAR system for interior layout and design. Insights were gathered as participants constructed physical kitchen mock-ups using cardboard boxes and designed SAR interfaces with office supplies, guided by a list of essential kitchen items. Before the tasks, participants were briefed at a high level on SAR's capabilities, limitations, and task instructions, and informed that the session would be video recorded. They were encouraged to articulate their vision of an ideal spatial design technology, addressing experiences, interactions, preferred input modalities, and desired SAR display outputs, including optimal UI placements without expert knowledge. All responses and processes were recorded for analysis. The Phi Coefficient was applied to find common trends within quantitative datasets.

1.2 Tasks

Participants were asked to complete four tasks:

Task 1: Observe participants' designing of the layout of their kitchen in a confined space, focusing on extending beyond the limitations of basic boxes. Within 15 minutes, participants used provided boxes to model cabinets, furniture, and appliances, while articulating their strategies for arranging boxes to achieve their desired designs (Fig.1: a).

Task 2: Observe participants' definition of the boxes or box stacks by utilizing sticky notes. Before affixing the labels, participants were asked to articulate their expectations regarding the SAR system's ability to identify and correlate the virtual objects to their physical cardboard counterparts. They had 5 minutes to complete the task (Fig.1: b).

Task 3: Participants were asked to design a SAR-based UI and elaborate how their designs could support them in augmenting and enhancing their space. The focus involved investigating user preferences for preferred interaction methods, as well as how a

¹ <https://www.cadwalk.global>

² <https://www.bigplansmelbourne.com.au/commercial-walkthrough-floorplans>

³ <https://www.huawei.com/>

⁴ <https://www.ikea.com/>

SAR user interface (their placement, size, and configuration of UI components) could best utilize the space available (Fig.1:c).

Task 4: Examined how participants interacted with a spatial computer system to perform tasks and functions (in this case, deletion) (Fig.1: c).

1.3 Data Collected

Data collected included demographic questionnaires, observations of participant behavior including problem solving, how participants configured and labelled objects, their preferences around how SAR could be utilized in the space (e.g., preferred surfaces for projecting UIs, participants' own UI designs, expected and preferred interaction modalities, and their overall expectations for the system.

4 RESULTS

1.1 Task 1 results: Spatial Configuration and Layout Design

Participants approached the task of configuring a compact kitchen with varying preferences for spatial organization. While many drew inspiration from their own kitchens for familiarity, others exhibited unique approaches. Common trends included favoring L-shaped and parallel galley layouts, with some incorporating elements like kitchen islands or shelving units for an immersive experience.

Participants repurposed materials such as cardboard boxes to mimic real-world features, stacking them to construct compartments, shelving, and even wall-mounted elements. Without significant constraints on how to use the boxes, some participants construct shelves and storage units, driven by their pursuit of an ideal kitchen layout. For instance, some participants turned boxes around to allow them to have hanging cabinets. One participant assembled cardboard pieces and smaller boxes to simulate a spice rack or stack two large boxes to represent an on-wall cabinet. While these constructions did not precisely mirror real-world objects, they effectively conveyed conceptual ideas.

1.2 Task 2 results: Object Definition Through Labelling or Semantic labelling

The results were influenced by participants' habits, perspectives, and personal approaches to problem-solving and convenience. During the task, some participants placed labels between boxes to combine boxes into single items, while others utilized labels to indicate the number of drawers, and even illustrated the direction of drawer openings. In addition, participants consistently positioned the sticky notes within the comfort of their arm's reach leading to an impression that the interface designs for labelling process should be interactive at eye level and situated within the users' peripersonal (reachable) space.

1.3 Task 3 results: Interactivity in Spatial Computing

Task 3 delves into the analysis of projected UIs regarding their locations, sizes and appearance. Participants explored various interaction modalities, favoring gestures, voice commands, and touch interfaces for spatial engagement then elaborated via their own crafted prototypes using office supplies with their own preference of paper size. Participants had no restrictions on the interfaces' placement or display configuration.

Many participants opted for two UI placement locations, with the front of the boxes being the most popular choice (selected 11 times), followed by the wall (8 times) and the top of the box (5 times), while only one selected the floor claiming ergonomic reasons. The Phi Coefficient analysis revealed a moderate association between the box's Top surface and Front surface conditions ($X^2 = 0.646$) and between the box's Top surface and Wall conditions ($X^2 = 0.573$). In contrast, the box's Front surface and

Wall conditions exhibited a weaker association ($X^2 = 0.331$), while the box's Top surface and other surfaces showed a minimal association ($X^2 = 0.12$). All remaining comparisons demonstrated negligible associations ($X^2 < 0.05$). These results suggested a partial alignment in user preferences for certain UI projection locations, particularly involving the Top surface, while other conditions displayed minimal or no correlation.

During the task, participants expressed a preference for intuitive interaction methods. Gestures were particularly preferred for their intuitive control, often paired with visual cues such as highlighted borders. Voice commands were also favored for their convenience, often paired with visual menus to enhance usability. Touch interfaces, modelled after familiar devices like tablets, were frequently utilized on accessible surfaces, complemented by responsive UI designs for flexibility. Some participants discussed advanced modalities, such as eye gaze tracking and Brain-Machine Interfaces. The Phi Coefficient only showed minimal associations among Voice interface vs Gestures ($X^2 = 0.1958374$) with visual cues, Touch interface vs Others ($X^2 = 0.2558711$), and Gestures with visual cues vs Touch interface ($X^2 = 0.1958374$).

Participants favored simple and functional UI sizes, opting to align the dimensions with A4 paper or tablet sizes. Phi Coefficient analysis revealed strong associations between "As large as the box (breadth)" and "Tablet/A4 size" ($X^2 = 0.678$) and between "Tablet/A4 size" and "Other sizes" ($X^2 = 0.723$), while the association between "Tablet/A4 size" and "Responsive" was weaker ($X^2 = 0.183$). These findings suggest a preference for combining smaller tablet-like interfaces for control and larger interfaces for detailed adjustments, driven by the impracticality of overly small elements. Some participants proposed a responsive approach, suggesting interfaces that adjust in size based on context or user needs, enhancing usability. Unique UX/UI concepts included zoom gestures for detailed views and modular menus for greater functionality.

Phi Coefficient analysis revealed strong associations in visual design preferences, particularly between Carousel and Gallery thumbnails ($X^2 = 0.803$), Carousel with filter and Gallery with filter ($X^2 = 0.606$), and Gallery with filter and other designs ($X^2 = 0.646$). In contrast, minimal associations were observed between Carousel with filter and others ($X^2 = 0.462$), Carousel and others ($X^2 = 0.331$), and Carousel and Carousel with filter ($X^2 = 0.409$). The remaining comparisons showed negligible associations ($X^2 < 0.05$). These findings indicate a preference for displaying inventory assets using Carousel and Gallery formats, with or without filters, reflecting a tendency among participants rely heavily on certain design aspects.

1.4 Task 4 results: Functional Interaction for Commands

Participants used intuitive methods such as gestures and voice controls to execute commands such as delete. Gesture-based actions, like grabbing and discarding virtual objects, often include symbolic actions like "crumbling" to signify deletion. Button-based interfaces, such as trash icons, were also preferred, with participants highlighting the importance of confirmation windows to avoid accidental deletions.

5 DISCUSSION

The findings reveal participants' inherent desire to replicate real-life experiences and prioritize usability in SAR systems. Most participants approached kitchen design with clear, preconceived layouts or relied on references to address spatial challenges, reflecting a strong preference for replicating familiar designs and environments. Their creative use of boxes to construct immersive layouts underscores the need for SAR systems to bridge the gap between symbolic representations and realistic visualizations.

Participants used boxes to approximate object dimensions and spatial volumes, showcasing their ability to conceptualize designs within given constraints. Nonetheless, this creative flexibility must be balanced with precision and functionality to ensure that designs align with industrial standards. Such alignment is crucial to minimize complications, reduce misinterpretations, and mitigate costs during the transition to construction as even small discrepancies in dimensions can result in inefficiencies and delays. While SAR systems currently offer features like real-time adjustments that enable users to test and refine designs dynamically, ensuring that early-stage prototypes closely resemble the final design is critical for streamlining the overall design and construction process.

Future SAR systems must prioritize adaptability, precision, and usability to accommodate diverse user needs. Incorporating flexible tools and interfaces that support intuitive adjustments will empower users to accurately approximate sizes, configurations, and spatial relationships. Key preferences such as gesture-based controls, voice commands, and responsive touch interfaces indicate that future systems should emphasize multi-modal interactions to align with natural user behaviors. Advanced features like adaptive UIs and augmented sensory feedback could further enhance the integration of physical and digital design elements, creating immersive environments. This study prioritized user-centric observations over technical evaluations of projected SAR. As mentioned before (**Introduction**), the focus was on understanding user behaviors and their interactions with SAR systems rather than delving into the technical aspects of SAR technology itself. The study aimed to observe how users engage with SAR interfaces to identify usability challenges and contextual problems that arise during usage, rather than evaluating or proposing advancements in SAR technology. For this reason, the study intentionally avoided specific configurations, such as whether the projection was 3D stereoscopic or not.

Novice users primarily concern about the visual clarity and functional convenience of SAR in designing their space, rather than its technical intricacies. This study highlights user engagement strategies and expectations, emphasizing customizable modular interfaces and real-time responsiveness while noting limitations in achieving precise spatial overlays. Future research should address these limitations to advance SAR UX/UI design and functionality in practical applications.

The study adopted an agnostic approach to SAR configurations, allowing us to explore user behavior and interactions broadly without tying the findings to specific technological setups or requirements. By concentrating on the user experience, this research seeks to inform design principles that address real-world usage needs rather than the technical evolution of SAR itself.

6 CONCLUSION

This study highlighted significant user preferences for input methods, projection locations, and UI design in world-building SAR systems. Gesture controls, especially with visual cues, were the most preferred input method when combined with voice commands and touch interfaces. Advanced input options, such as brain signal inputs or integration with familiar devices like tablets, were anticipated as promising future innovations, alongside some interest in tactile controllers and eye-gaze controls. For projection locations, the front surface of the box, the wall, and the top of the box were deemed most suitable, with little interest in using the floor. In terms of UI design, carousel layouts were the most favored, followed by gallery thumbnails, with or without filter, and unconventional styles. Buttons were preferred for specific functions like deletion, though gestures were also suggested. Based on these findings, gestures, voice, and touch inputs were prioritized, alongside the identified projection surfaces and UI

design patterns, to guide the development and refinement of SAR systems for collaborative spatial design.

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